Trainings

General Information

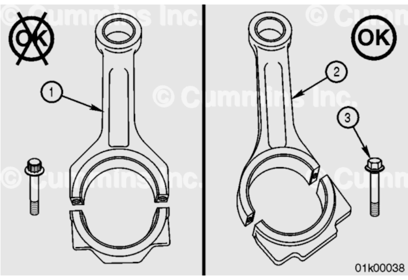
Note : There are two different piston and liner combinations available for this engine platform. The ferrous cast ductile (FCD) piston **must** be used in combination with a nitrided cylinder liner. The steel piston **must** be used in combination with a high boron liner. It is important that these parts are used in this combination stated and **never** mixed. An engine **must never** be run with a mixture of FCD and steel pistons. It **must** contain **only** a full set of FCD pistons or a full set of steel pistons. The liners can be distinguished from each other by appearance. The nitride liners will have a matte appearance on the outer circumference. The high boron liners are bright and shiny in appearance.

Install

Use the following steps for engines containing angle split connecting rods (2) **only**.

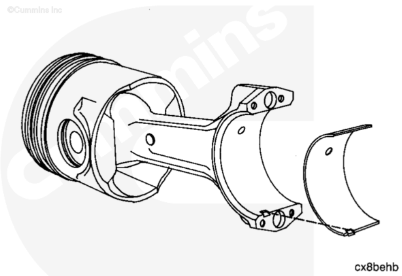
Use the following procedure for engines containing straight split connecting rods (1). Refer to Procedure 001-045 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html)

Note : The angle split connecting rod (2) can be identified by 6 point head capscrews (3) and the angled joint



Use a lint-free cloth to clean the connecting rod and the bearing shells.

Install the connecting rod bearing. Make sure the tang is positioned as shown. The end of the bearing **must** be even with the cap mounting surface.



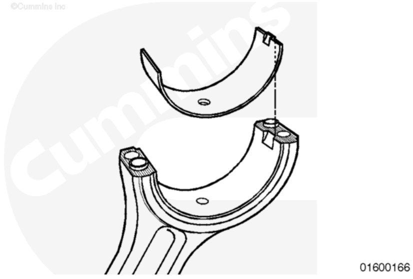
⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the backside of the bearing shells.

Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086.

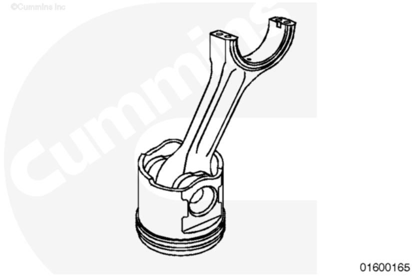
The bearings **must** be installed in their original location, if new bearings are **not** used.

Note : All of the rod bearings are identical.



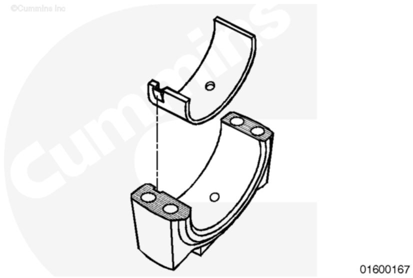
The guides aid the assembly and protect the crankshaft.

Install two connecting rod guides, or equivalent, in the rod.



⚠ CAUTION ⚠

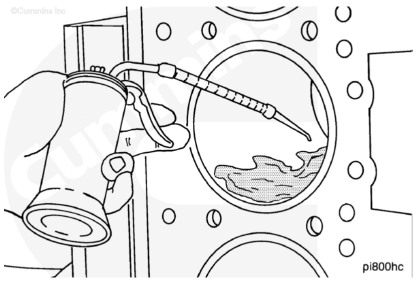
The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if they are mixed.



Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

Lubricate the bearing surface with Lubriplate™ 105, Part Number 3163086.

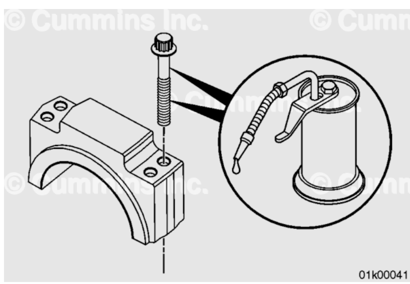
Lubricate the cylinder liner with clean engine oil. The entire bore **must** be lubricated.



Lubricate the connecting rod capscrews head flanges with clean Society of Automotive Engineers (SAE) W140 gear oil.

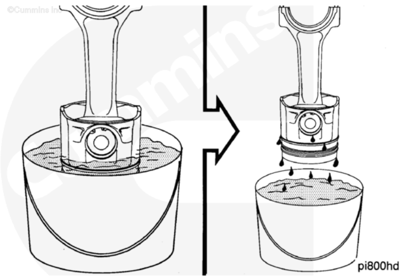
Lubricate the capscrew threads with clean engine oil.

Install the washers and capscrews in the caps.

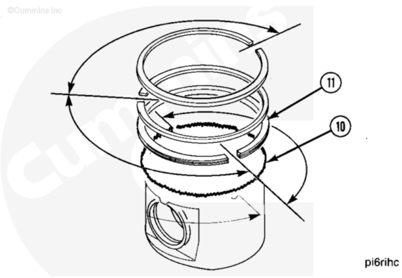


Immerse the piston in engine oil until the rings are covered.

Allow the excess oil to drip off the assembly.

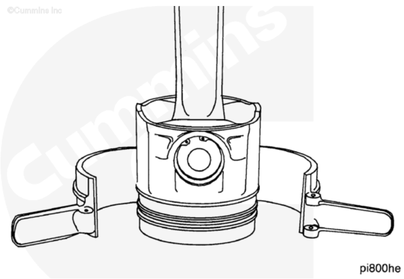


Check to make sure the ring gap position is still correct.



⚠ CAUTION ⚠

To reduce the possibility of engine damage, make sure the rings fit correctly in the piston.

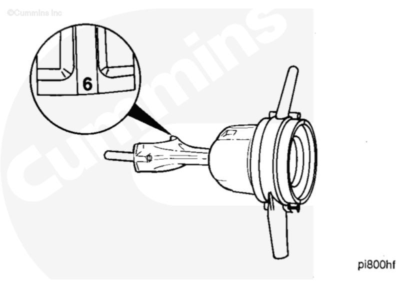


The ring compressor has a tapered bore. The small end of the taper **must** be positioned toward the piston skirt.

Use a piston ring compressor, Part Number 3375342, or equivalent. Install the ring compressor on the piston.

⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.



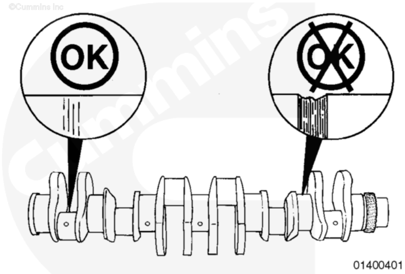
⚠ CAUTION ⚠

The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Rotate the crankshaft until the journal for the connecting rod being installed is at top dead center and centered in the cylinder bore.

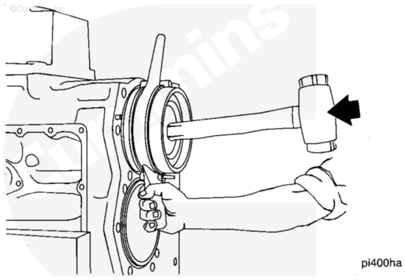
Note : The crankshaft is shown out of the engine for clarity.

Check the crankshaft rod journals for damage.



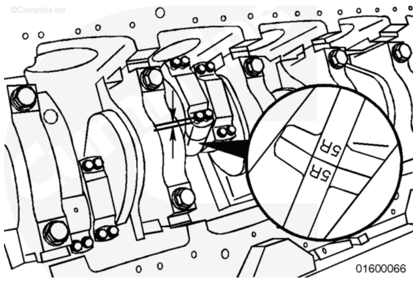
Install the connecting rod and piston until the ring compressor touches the cylinder block. Align the rod with the crankshaft journal.

Hold the ring compressor firmly against the block. Use a wooden hammer handle to push the piston into the liner.



⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same, damage to the engine can occur if the rods and rod caps are mixed.



⚠ CAUTION ⚠

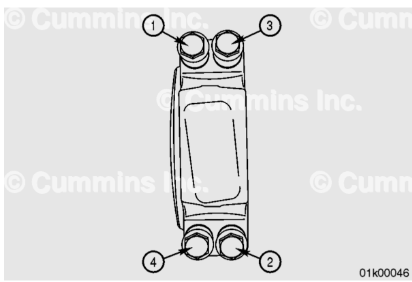
The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backward, engine damage will occur.

Install the connecting rod cap. Tighten the capscrews alternately and evenly to pull the cap over the dowel pins.

Use the following steps and sequence to tighten the capscrews.

Use an electronic torque wrench to tighten the connecting rods capscrews.

Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the graphic.



Torque Value:

1. 85 n•m [63 ft-lb]
2. Repeat step 1

3. Loosen all capscrews 2 revolutions minimum
4. 85 n•m [63 ft-lb]
5. Repeat step 4
6. Advance 60 degrees

If the final torque falls outside the range of 130 to 165 N•m [96 to 122 ft-lb], repeat steps 3 through 6.

Use the following torque procedure if accessing the connecting rod cap through the handhole cover if the torque angle dial is inaccessible.

Prior to applying this process, be sure that all capscrews have been tightened by hand as far as possible to prevent thread damage or mis-engagement.

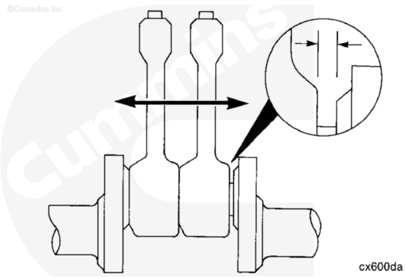
Use the following steps and sequence to tighten the capscrews.

Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the graphic.

Torque Value:

1. 85 n•m [63 ft-lb]
2. Repeat step 1
3. Loosen all capscrews two revolutions minimum
4. 85 n•m [63 ft-lb]
5. Repeat step 4
6. 140 n•m [103 ft-lb]
7. Repeat Step 6

Check the side clearance between the connecting rod and the crankshaft. Refer to Procedure 001-005 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-005-tr.html)



The connecting rod **must** move freely from side to side.

Connecting Rod and Crankshaft Side Clearance		
mm		in
0.30	MIN	0.012
0.52	MAX	0.020